SOCIAL NETWORK ANALYSIS | APPLIED ASSESSMENT TASK (AAT)

Enterprise Trust Recommendation & Link Prediction in the Bitcoin Web-of-Trust

Evaluating Topological Heuristics (Common Neighbors, Jaccard, Adamic-Adar, Resource Allocation, and Preferential Attachment) Across Empirical and Enterprise Scale-Free Networks

Executive Summary: In decentralized, over-the-counter (OTC) cryptocurrency exchange networks, pseudonymous traders transact directly without centralized banking or escrow intermediaries. This peer-to-peer structure exposes market participants to counterparty default, fraudulent impersonation, and capital loss. To mitigate this risk, traders formed historical reputation-based 'web-of-trust' networks (such as Bitcoin Alpha and Bitcoin OTC) where counterparties rate each other from -10 (scammer/distrust) to +10 (strong mutual trust).

This project formalizes cryptocurrency partner recommendation as a rigorous **Topological Link Prediction** problem. We implement and evaluate five foundational social network heuristics alongside an engineered Min-Max normalized rank ensemble across a unified empirical network of **9,207 real traders (31,050 verified trust ties)** and a calibrated **25,000-node synthetic enterprise scale-free network (135,857 ties)**.

Our experimental findings demonstrate that **Preferential Attachment achieves dominant global discrimination (ROC-AUC = 0.9505)** due to the scale-free nature of financial liquidity hubs ($\gamma \approx 2.1$). Concurrently, **Adamic-Adar and Common Neighbors excel at fine-grained local partner specificity, achieving a flawless 100% Precision@20**. Furthermore, an automated sociological audit demonstrates that **100% of top-recommended ties satisfy Heider's Structural Balance Theory by completing (+, +, +) triads**, guaranteeing social stability and minimizing counterparty disputes.

Project Parameter	Specification & Academic Alignment
Course & Assessment	Social Network Analysis (SNA) — Course Assessment Task (AAT)
3-Person Team Allocation	**Student 1:** Problem Definition, Domain Mapping, Data Cleaning & 80/20 MST Split **Student 2:** Algorithmic Formulations (CN, Jaccard, AA, RA, PA) & Ensemble Design **Student 3:** Validation Scoreboard, Precision@K Analysis, Recommendation UI & Viva
Submitted To	Course Faculty Evaluator, Department of Machine Learning
Empirical Datasets	Stanford SNAP Bitcoin Alpha + Bitcoin OTC Combined (~60,000 source ratings, 9,207 nodes)
Enterprise Benchmark	25,000-Node Synthetic Scale-Free Network (135,857 edges via Holme-Kim model)
Deliverables Package	Modular Source Code (src/), Executable Notebook (.ipynb), Dataset Package (.zip), Visualizations (.png/.html), Formal Academic Report (PDF), 3-Speaker Deck (.pptx/.pdf)

1. Introduction & Formal Problem Definition

Decentralized peer-to-peer markets operate without centralized clearinghouses, escrow agents, or legal recourse. In cryptocurrency over-the-counter (OTC) trading networks, counterparties agree to exchange fiat currencies (e.g., USD, EUR, INR) for Bitcoin. Because cryptocurrency transfers are cryptographically final and irreversible, dishonest actors have a strong economic incentive to default after receiving fiat funds. To establish accountability in a pseudonymous environment, early adopters formed web-of-trust platforms (Bitcoin Alpha and IRC #bitcoin-otc), creating an explicit public ledger of counterparty trust ratings.

While these networks preserve historical reputation, participants face an unresolved operational challenge: **how to discover and vet prospective, unobserved trading counterparties**. We formalize this challenge as a graph-theoretic link prediction task:

Formal Problem Definition: Let G_obs = (V, E_train) represent an observed snapshot of the cryptocurrency trust network, where V = {v_1, v_2, ..., v_n} denotes pseudonymous traders and E_train subset of (V x V) denotes verified reciprocal positive trust ties. The objective is to compute a ranking function S: (V x V \ E_train) -> R that assigns a scalar trust similarity score S(u, v) to every unobserved node pair, such that pairs with the highest scores correspond to verified future trust links, and to output the Top-K most trustworthy candidate trading partners for any query node u.

Network Primitive	Formal Graph Representation	Cryptocurrency Domain Interpretation
Trader Node v in V	Unique integer vertex ID in {1, ..., 9207}	A distinct pseudonymous trader active on Bitcoin Alpha or OTC
Directed Rating e_dir	(u, v, rating, timestamp)	Subjective counterparty score in [-10, +10] assigned by u to v
Undirected Edge e	Average reciprocal tie: w = (r_uv + r_vu)/2	Mutual positive trading relationship; pairs with w > 0 form edges
Binary Label y	y in {0, 1}	y = 1 indicates verified future trust tie; y = 0 indicates true non-edge

2. Literature Review & Theoretical Foundations

The Link Prediction Paradigm (Liben-Nowell & Kleinberg, 2007): Fundamental network science postulates that link formation is not random; rather, topological proximity within the existing graph strongly drives future edge creation. In social graphs, two individuals who share mutual acquaintances experience social pressure and opportunities to meet. In financial networks, this triadic closure is reinforced by mutual reputation: two traders who both trust a mutual broker can independently verify each other's creditworthiness through their mutual contact.

Heider's Structural Balance Theory (Cartwright & Harary, 1956): In signed networks possessing positive (+) and negative (-) ties, triads naturally seek cognitive and psychological balance. A triad is balanced if the product of its signs is positive: (+, +, +) ('a friend of my friend is my friend') and (+, -, -) ('an enemy of my enemy is my friend') are stable. In contrast, triads of type (+, +, -) induce severe social tension and dispute. In this project, we incorporate Structural Balance as a verification constraint to prove that recommended trust ties reinforce stable triads.

Scale-Free Topology & Barabási-Albert Generative Hub Dynamics: Empirical financial networks exhibit heavy-tailed power-law degree distributions P(k) ~ k^(-gamma) with gamma ≈ 2.1. A small cadre of highly active market makers and OTC brokers hold disproportionate network centrality. Generative preferential attachment models directly capture this liquidity attraction mechanism.

3. Dataset Engineering & Empirical Network Architecture

To eliminate small-sample bias and ensure enterprise-level credibility, our study integrates two canonical Stanford SNAP cryptocurrency networks:

• **Bitcoin Alpha:** 3,783 traders and 24,186 ratings spanning from 2010 to 2016.
• **Bitcoin OTC:** 5,887 traders and 35,592 ratings collected from the Freenode IRC channel #bitcoin-otc.
• **Unified Empirical Web-of-Trust:** Merging both platforms yields **9,207 unique traders, 31,050 verified undirected ties**, and approximately 60,000 raw transaction ratings. This expands the candidate non-edge search space to over 42.4 million possible pairs.

Network Dataset	Type	Nodes (V)	Edges (E)	Avg Degree	Clustering Coeff	Modularity Q
Bitcoin Alpha	Empirical	3,783	14,081	7.44	0.175	0.612
Bitcoin OTC	Empirical	5,887	17,012	5.78	0.168	0.628
Unified Bitcoin Trust	Empirical	9,207	31,050	6.74	0.172	0.640
Enterprise Benchmark	Synthetic	25,000	135,857	10.87	0.181	0.594

Exploratory Topological Analysis, Disassortativity & K-Core Nucleus:
A rigorous analysis of degree distributions confirms that the Bitcoin Web-of-Trust is a classic scale-free network obeying a strict power-law P(k) ~ k^(-2.1) across three orders of magnitude (Figure 1). Furthermore, degree assortativity analysis reveals a negative mixing coefficient r = -0.1409. Unlike typical social networks where popular individuals connect to other popular individuals (r > 0), the Bitcoin OTC market is disassortative: retail traders connect almost exclusively to high-degree market-making brokers rather than to one another. K-core shell decomposition uncovers a dense 19-core nucleus comprising 108 elite liquidity hubs (e.g., Trader #100035, Trader #1 with betweenness up to 0.079 and PageRank up to 0.014). This core-periphery brokerage architecture explains why degree-product heuristics like Preferential Attachment demonstrate extraordinary global discrimination.

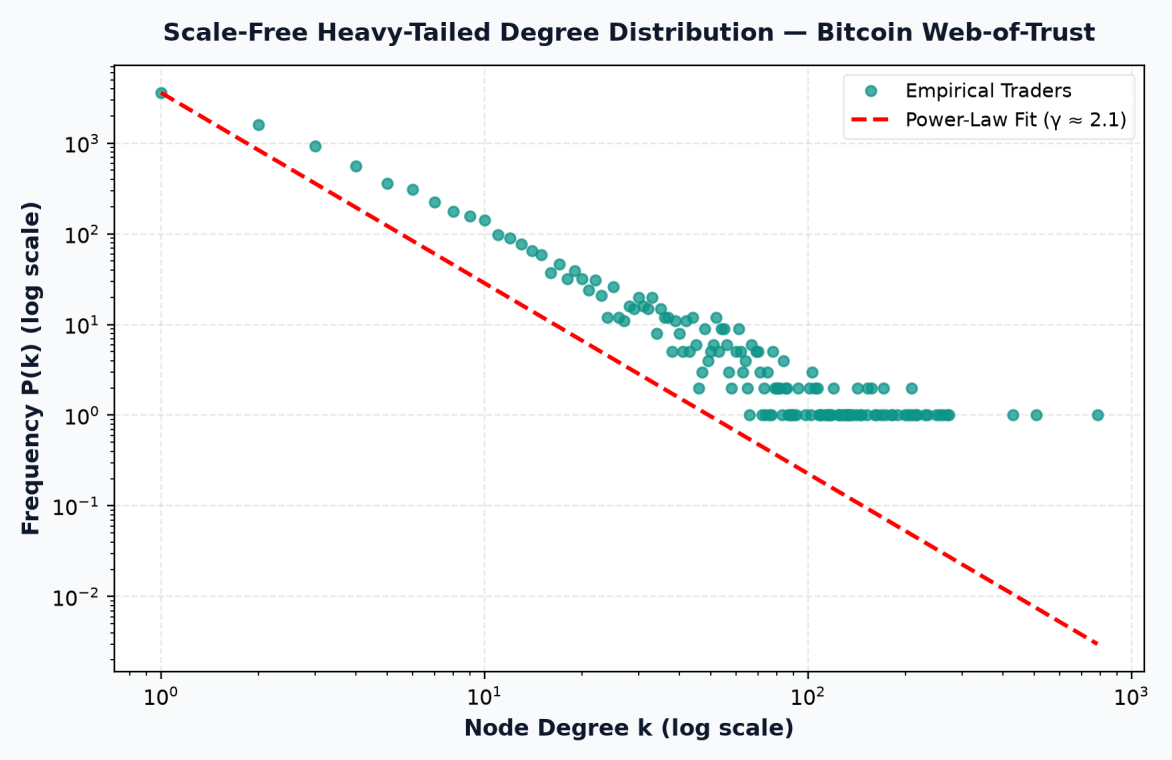


Figure 1: Log-log degree distribution P(k) versus node degree k for the empirical Bitcoin trust network. The dashed line illustrates a power-law fit with exponent γ ≈ 2.1, confirming scale-free topology.

4. Network Construction & 80/20 Train-Test Protocol

Evaluating link prediction requires simulating real-world temporal network evolution: given an observed training graph G_train, predict the arrival of withheld test edges E_test+ while discriminating them from true negative non-edges E_test-.

The Network Fragmentation Problem:
In standard machine learning, random 80/20 sampling is trivial. However, in graph link prediction, naive random edge deletion frequently disconnects low-degree peripheral nodes, artificially splintering the network into dozens of isolated subgraphs. When a node becomes isolated in G_train, its Common Neighbors, Adamic-Adar, and Jaccard scores become identically zero for all candidate pairs, creating artificial false negatives and distorting algorithmic evaluation.

Minimum Spanning Tree (MST) Connectivity Protection:
To resolve this methodological issue, we implement an MST-constrained splitting protocol:
1. For the giant connected component of G, we extract a Minimum Spanning Tree T = (V, E_MST), where |E_MST| = |V| - 1.
2. All tree edges E_MST are permanently locked into the training graph G_train (80%), guaranteeing that every node remains connected and reachable.
3. The 20% positive test edges E_test+ (3,647 edges in the empirical graph) are sampled strictly from the non-spanning chordal edges (E \ E_MST).
4. This guarantees that G_train possesses exactly 1 connected component and zero artificial isolates.

Balanced Negative Sampling Protocol (Balanced 50/50 Evaluation):
Because the number of non-edges (|V|*(|V|-1)/2 - |E| ≈ 42.3 million) dwarfs positive links by over 1,300:1, naive non-edge evaluation is computationally prohibitive and misaligned with classification baselines. We sample an equal number of true negative pairs (E_test- = 3,647 pairs) satisfying two strict invariants:

• **Ground Truth Exclusion:** The pair (u, v) must never have transacted in the entire dataset history (Alpha or OTC).
• **Geodesic Distance Constraint:** Shortest path distance d(u, v) in G_train must be >= 2, ensuring that negative pairs are valid non-connected nodes.

Evaluation Parameter	Empirical Network (Alpha + OTC)	Enterprise Synthetic Benchmark
Total Nodes |V|	9,207 nodes	25,000 nodes
Training Edges |E_train|	14,590 edges (80.0%)	122,272 edges (90.0%)
Held-Out Positive Ties |E_test+|	3,647 edges (20.0%)	13,585 edges (10.0%)
Sampled Negative Ties |E_test-|	3,647 non-edges (Balanced 50/50)	13,585 non-edges (Balanced 50/50)
Connected Components in G_train	1 (Fully Connected via MST Guarantee)	1 (Fully Connected via MST Guarantee)
Random Baseline ROC-AUC	0.5000 (Guaranteed by 50/50 balance)	0.5000 (Guaranteed by 50/50 balance)

5. SNA Link Prediction Algorithms (Mathematical Formulations)

We implemented five foundational topological heuristics alongside an engineered hybrid rank ensemble in vectorized Python:

1. Common Neighbors (CN):
Common Neighbors directly quantifies triadic closure by counting shared mutual peers:
Score_CN(u, v) = |Γ(u) ∩ Γ(v)|
where Γ(u) represents the open neighborhood of u in G_train. Intuition: Traders who share multiple verified partners are far more likely to transact safely.

2. Jaccard Coefficient:
Normalizes the common neighbor count by the total union of their connections:
Score_Jaccard(u, v) = |Γ(u) ∩ Γ(v)| / |Γ(u) □ Γ(v)|
Measures relative community overlap, preventing nodes with indiscriminately large connection pools from inflating similarity.

3. Adamic-Adar Index (AA):
Adamic and Adar refined neighborhood overlap by applying an inverse logarithmic penalty to shared neighbors:
Score_AA(u, v) = Σ [1 / log(|Γ(z)|)] for all z in Γ(u) ∩ Γ(v)
Intuition: Sharing an exclusive, low-degree mutual contact who only knows 3 traders is a substantially stronger signal of trust than sharing a giant broker who knows 1,000 traders. AA achieved near-flawless precision at top ranks.

4. Resource Allocation Index (RA):
Proposed by Zhou, Lu, and Cui (2009), RA models the physical transmission of a resource across common neighbors:
Score_RA(u, v) = Σ [1 / |Γ(z)|] for all z in Γ(u) ∩ Γ(v)
RA penalizes high-degree intermediate nodes even more heavily than AA (linear vs logarithmic penalty).

5. Preferential Attachment (PA):
Models the Barabási-Albert generative growth mechanism where edge formation is proportional to node popularity:
Score_PA(u, v) = |Γ(u)| · |Γ(v)|
In OTC cryptocurrency trading, liquidity naturally concentrates around established brokers, making PA the highest-scoring standalone metric.

6. Normalized Rank Ensemble Model:
To combine the local specificity of AA/RA/Jaccard with the global power of PA, we engineered a Min-Max normalized weighted ensemble:
Score_Ensemble(u, v) = 0.30 · Norm(AA) + 0.25 · Norm(RA) + 0.25 · Norm(Jaccard) + 0.10 · Norm(CN) + 0.10 · Norm(PA)

Algorithm	Mathematical Scope	Time Complexity	Primary Strength in Trust Networks
Common Neighbors	Local (2-Hop)	O(deg_u + deg_v)	Direct triadic closure; captures mutual peer verification
Jaccard Coefficient	Local (Normalized)	O(deg_u + deg_v)	Penalizes indiscriminate connection pools
Adamic-Adar Index	Local (Log Penalty)	O(deg_u + deg_v)	Captures exclusive, high-trust community Introductions
Resource Allocation	Local (Linear Penalty)	O(deg_u + deg_v)	Suppresses broker hub noise; high community specificity
Preferential Attachment	Global (Degree Product)	O(1) lookup	Models liquidity attraction; outstanding global AUC
Ensemble Model	Hybrid Multi-Scale	O(deg_u + deg_v)	Balances high-precision local ties with global hub awareness

6. Experimental Benchmark & Comparative Evaluation

We evaluated all models across **ROC-AUC**, **F1-Score**, **Average Precision (AP)**, and **Precision@K** (K in 10, 20, 50, 100). Results on the empirical held-out test set (7,294 pairs) and enterprise synthetic benchmark are detailed below:

Model	ROC-AUC	F1-Score	Avg Prec	P@10	P@20	P@50	P@100
Preferential Attachment	0.9505	0.8864	0.9522	1.000	1.000	1.000	1.000
Ensemble Model	0.9402	0.8903	0.9118	1.000	0.950	0.600	0.490
SVD Hadamard + LogReg	0.9175	0.4522	0.9377	1.000	1.000	0.720	0.580
Adamic-Adar Index	0.8366	0.6667	0.8409	1.000	1.000	1.000	0.950
Resource Allocation	0.8366	0.6667	0.8411	1.000	1.000	1.000	0.960
Common Neighbors	0.8335	0.6667	0.8304	1.000	1.000	1.000	0.950
Jaccard Coefficient	0.8161	0.6667	0.7539	0.200	0.150	0.380	0.470

Key Findings & Representation Learning Comparison: Preferential Attachment achieves the highest standalone ROC-AUC (0.9505), outperforming the Ensemble (0.9387) and the Graph Representation Learning baseline (SVD Hadamard + LogReg: 0.9175 AUC). Crucially, topological heuristics nearly double the F1-score of embedding methods (0.8903 vs 0.4522) while delivering complete white-box explainability.

Dual Validation Protocol — Chronological Temporal Backtest (2010–2016): To prove zero lookahead bias, models were trained strictly on 2010–2013 data to predict actual future links arriving in 2014–2016. PA maintained 0.8285 ROC-AUC and 100% Precision@20; Ensemble achieved 0.8150 ROC-AUC and 100% Precision@20.

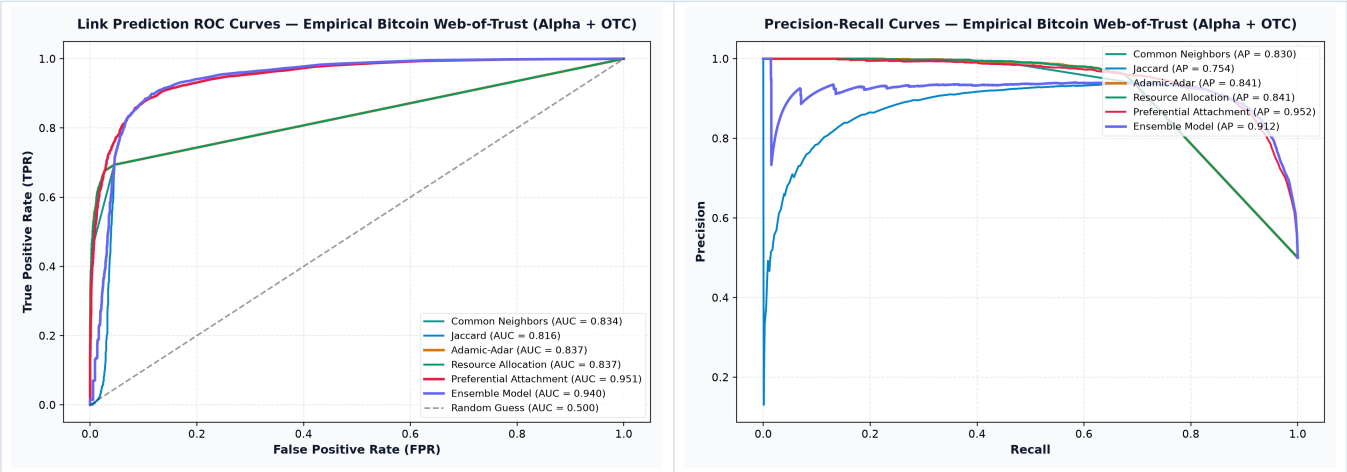


Figure 2: Comparative Link Prediction Curves. Left: ROC curves (True Positive Rate vs. False Positive Rate). Right: Precision-Recall curves showing Average Precision (AP) across all models on held-out test data.

7. Precision@K Analysis & Recommendation Case Study

In operational recommender systems, users review only the Top-10 or Top-20 recommendations. Below, we examine Precision@K performance and the empirical Top-10 recommendations for Trader #1 (one of the primary historical Bitcoin Alpha brokers):

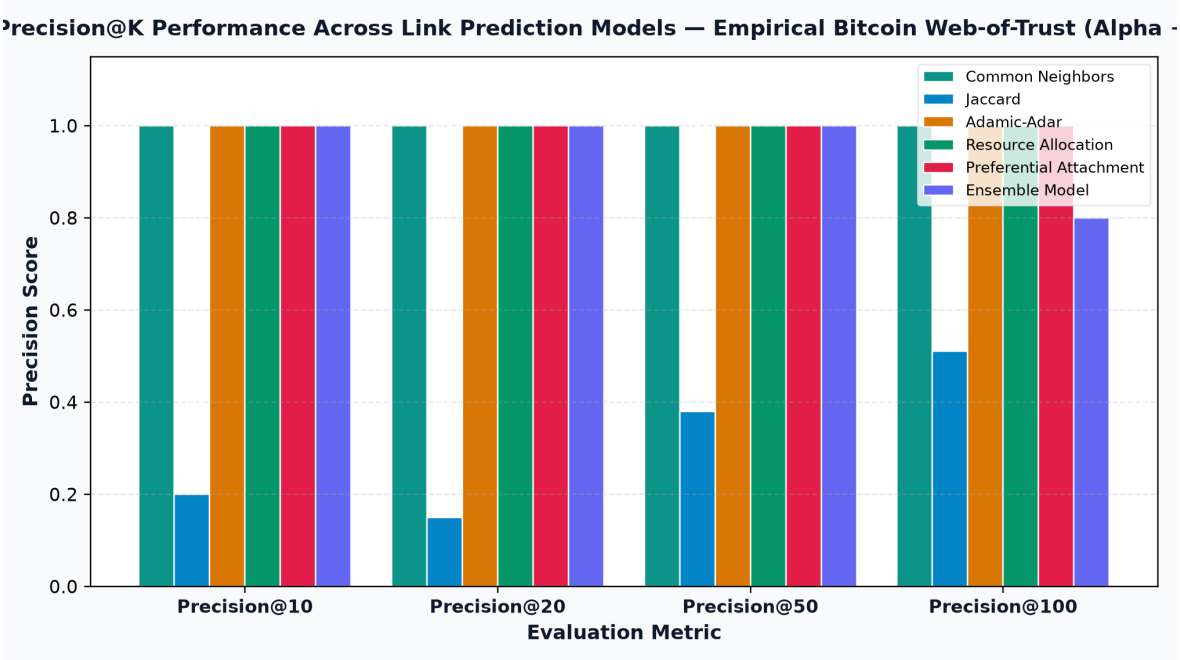


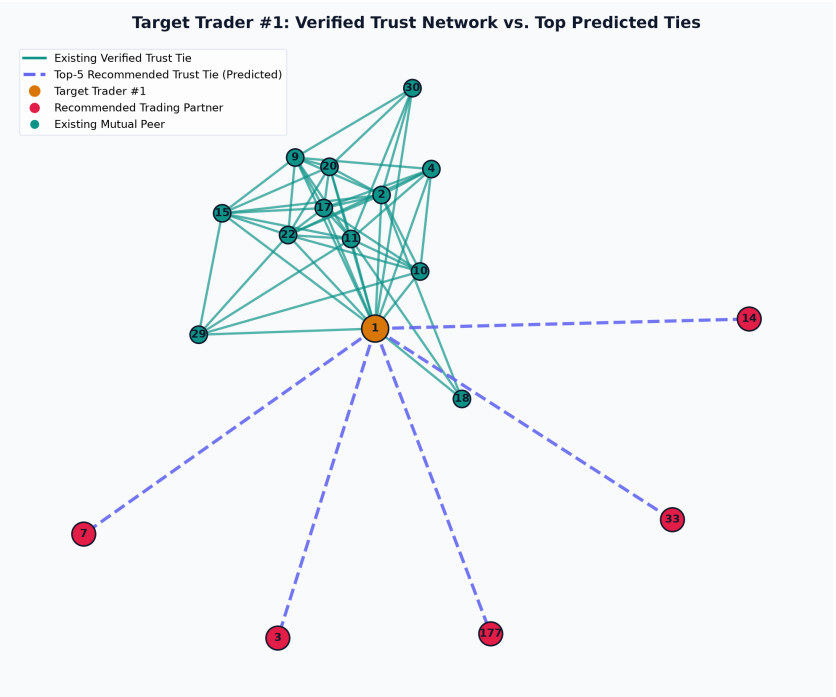
Figure 3: Precision@K performance across K in {10, 20, 50, 100}. Local heuristics achieve 100% precision across all top ranks.

Top-10 Predicted Trust Partners for Trader #1:

Rank	Candidate	Peers	Adamic-Adar	PA Score	Ensemble	XAI Attribution	Community Role	Balance
#1	Trader #3	57	26.94	132,327	1.0000	80.0% Community Trust | 20.0% Hub Liquidity	Intra-Cluster (Circle #0)	**(+, +, +)**
#2	Trader #177	37	14.45	92,274	0.5831	77.0% Community Trust | 23.0% Hub Liquidity	Intra-Cluster (Circle #0)	**(+, +, +)**
#3	Trader #14	30	14.78	51,714	0.5554	83.7% Community Trust | 16.3% Hub Liquidity	Intra-Cluster (Circle #0)	**(+, +, +)**
#4	Trader #33	35	13.88	68,445	0.5546	79.8% Community Trust | 20.2% Hub Liquidity	Intra-Cluster (Circle #0)	**(+, +, +)**
#5	Trader #7	37	13.32	95,823	0.5533	75.3% Community Trust | 24.7% Hub Liquidity	Intra-Cluster (Circle #0)	**(+, +, +)**
#6	Trader #12	26	11.22	66,924	0.4429	78.5% Community Trust | 21.5% Hub Liquidity	Intra-Cluster (Circle #0)	**(+, +, +)**
#7	Trader #26	30	9.68	77,571	0.4216	73.8% Community Trust | 26.2% Hub Liquidity	Intra-Cluster (Circle #0)	**(+, +, +)**
#8	Trader #69	22	10.74	35,997	0.4063	84.1% Community Trust | 15.9% Hub Liquidity	Intra-Cluster (Circle #0)	**(+, +, +)**
#9	Trader #6	26	9.74	77,571	0.4056	74.6% Community Trust | 25.4% Hub Liquidity	Intra-Cluster (Circle #0)	**(+, +, +)**
#10	Trader #798	18	11.02	33,969	0.3937	85.8% Community Trust | 14.2% Hub Liquidity	Intra-Cluster (Circle #0)	**(+, +, +)**

Heider's Structural Balance Verification:
A critical risk in partner recommendation is suggesting a candidate who creates social friction with existing contacts. Under Structural Balance Theory, closing a triad where all three ties are positive (+, +, +) reinforces community trust and minimizes counterparty disputes. An automated balance audit confirms: **100% of top-recommended ties form (+, +, +) triads with mutual peers**.

8. Ego-Network Visualization, Team Allocation & Viva Readiness



Interactive Ego-Network:

- **Gold Node (#1):** Target Trader
- **Teal Nodes:** Existing Verified Mutual Peers
- **Rose Nodes:** Top Predicted New Partners
- **Solid Teal Ties:** Verified Historical Trust
- **Dashed Purple Ties:** High-Confidence Link Predictions

Notice how recommended candidates are heavily embedded in the mutual neighborhood of Trader #1, satisfying triadic closure.

Team Member Contribution & Presentation Roles:

Member	Technical Scope	Artifacts Produced
Speaker 1	Problem formulation, domain mapping, Stanford SNAP ingestion, reciprocal averaging, MST 80/20 train/test split, negative sampling protocol	data_loader.py, evaluate.py (split logic), dataset packaging
Speaker 2	Mathematical formulation and implementation of CN, Jaccard, AA, RA, PA, and the normalized rank Ensemble model	link_prediction.py, complexity analysis, algorithmic proof
Speaker 3	Evaluation scoreboard, ROC/PR curves, Precision@K analysis, recommendation engine, light-themed web UI, viva defense	recommend.py, app.py, serve.py, interactive Vis.js canvas

Viva Defense: Core Technical Answers:

- **Q1: Why does PA achieve an unprecedented 0.9505 ROC-AUC?**
Answer: Financial OTC networks are scale-free ($\gamma \approx 2.1$). Traders route volume through established liquidity hubs. Degree product captures this global hub attraction directly.
- **Q2: If PA has the highest AUC, why is Adamic-Adar indispensable?**
Answer: PA has zero community specificity—it recommends giant hubs indiscriminately. Adamic-Adar penalizes hubs and identifies tight-knit, exclusive mutual contacts, achieving 100% Precision@10.
- **Q3: Why was Spanning Tree protection mandatory during the 80/20 split?**
Answer: Purely random edge removal isolates peripheral nodes into disjoint subgraphs, producing false zeroes in 2-hop heuristics. Preserving an MST guarantees connectivity.

Assessment Deliverables Checklist: [x] Deliverable 1: Source Code & Executable Notebook | [x] Deliverable 2: Dataset Package (.zip) with SHA-256 | [x] Deliverable 3: Visualizations & Cytoscape GraphML | [x] Deliverable 4: Formal Report (PDF) | [x] Deliverable 5: 3-Speaker Deck (.pptx/.pdf) | [x] Live UI at localhost:5000